

NeuroTrax SensorDB CAVIA

Project overview: cognitive assessments + large-scale smartphone SQL digital phenotyping

A growing pipeline for a rich digital biomarker model: baseline phenotype and T1-to-T2 change phenotype.

Data Scale First

Raw data scale

10 TB

approximate project storage scale

Approx. per patient

122 GB

10 TB divided across NeuroTrax patients

Patient and NeuroTrax Overview

Patients total

83

Patients with T1

83

Patients with global delta

62

Cognitive patients with device label

82

Phase 2 Work Underway

Target analysis tables

21

Reviewed table files

10

Selected features

29

Two Main Project Outcomes

1

T1 baseline digital phenotype

Describe smartphone-derived behavior/context around the first cognitive assessment and later test prediction of NeuroTrax T1 score.

2

T1-to-T2 digital phenoc

Quantify digital changes between T1 and T2 to help predict NeuroTrax T2 cognitive

Experiment Time Span

First T1

2025-01-05

Last T2

2025-12-30

NeuroTrax Feature Domains

Overall / Global global_T1, global_T2, global_delta	Memory memory_T1, memory_T2, memory_delta	Executive function ef_T1, ef_T2, ef_delta
Processing speed processing_speed_T1/T2/delta	Verbal verbal_T1, verbal_T2, verbal_delta	Motor motor_T1, motor_T2, motor_delta